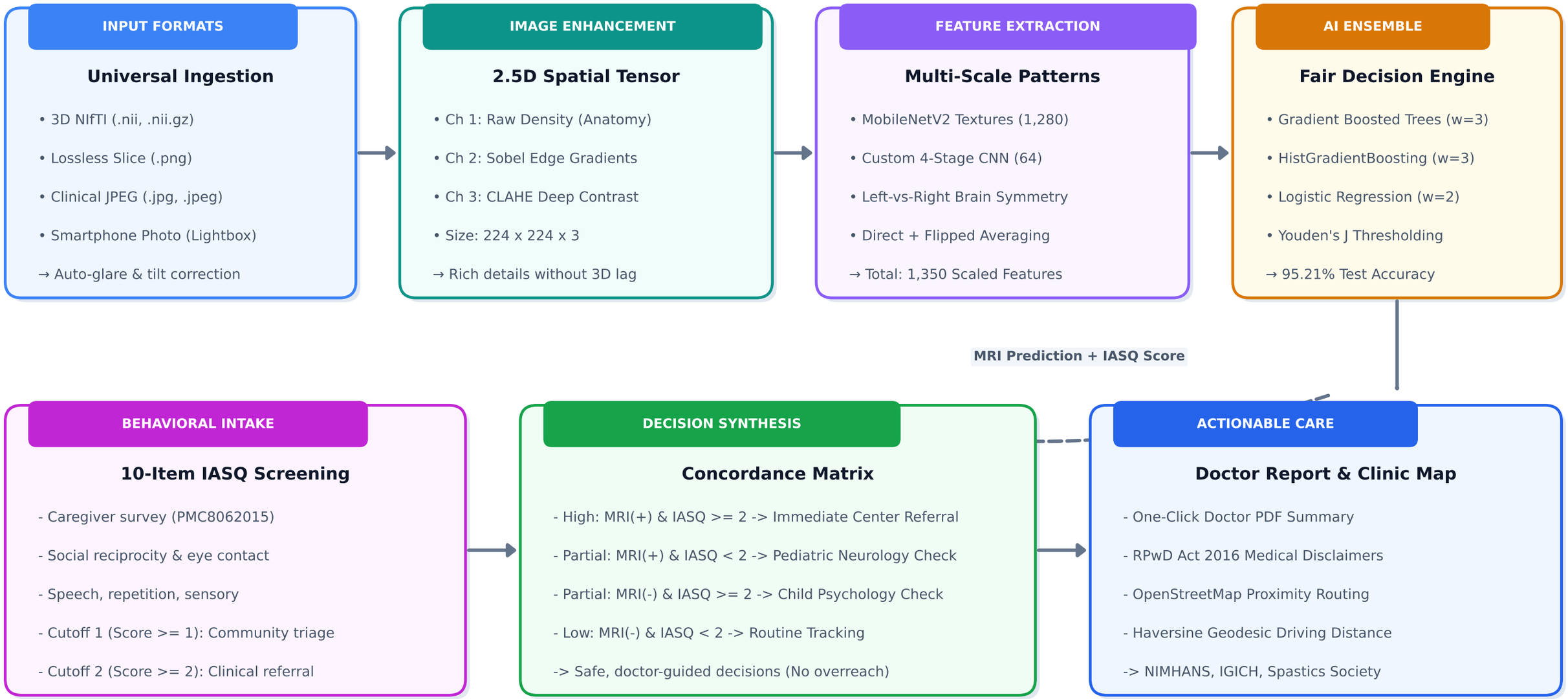
INPUT FORMATS
Universal Ingestion
• 3D NIfTI (.nii, .nii.gz)
• Lossless Slice (.png)
• Clinical JPEG (.jpg, .jpeg)
• Smartphone Photo (Lightbox)
→ Auto-glare & tilt correction
IMAGE ENHANCEMENT
2.5D Spatial Tensor
• Ch 1: Raw Density (Anatomy)
• Ch 2: Sobel Edge Gradients
• Ch 3: CLAHE Deep Contrast
• Size: 224 x 224 x 3
→ Rich details without 3D lag
FEATURE EXTRACTION
Multi-Scale Patterns
• MobileNetV2 Textures (1,280)
• Custom 4-Stage CNN (64)
• Left-vs-Right Brain Symmetry
• Direct + Flipped Averaging
→ Total: 1,350 Scaled Features
AI ENSEMBLE
Fair Decision Engine
• Gradient Boosted Trees (w=3)
• HistGradientBoosting (w=3)
• Logistic Regression (w=2)
• Youden's J Thresholding
→ 95.21% Test Accuracy
MRI Prediction + IASQ Score
BEHAVIORAL INTAKE
10-Item IASQ Screening
- Caregiver survey (PMC8062015)
- Social reciprocity & eye contact
- Speech, repetition, sensory
- Cutoff 1 (Score >= 1): Community triage
- Cutoff 2 (Score >= 2): Clinical referral
DECISION SYNTHESIS
Concordance Matrix
- High: MRI(+) & IASQ >= 2 -> Immediate Center Referral
- Partial: MRI(+) & IASQ < 2 -> Pediatric Neurology Check
- Partial: MRI(-) & IASQ >= 2 -> Child Psychology Check
- Low: MRI(-) & IASQ < 2 -> Routine Tracking
-> Safe, doctor-guided decisions (No overreach)
ACTIONABLE CARE
Doctor Report & Clinic Map
- One-Click Doctor PDF Summary
- RPwD Act 2016 Medical Disclaimers
- OpenStreetMap Proximity Routing
- Haversine Geodesic Driving Distance
-> NIMHANS, IGICH, Spastics Society